ADITYA PRANAV SHAH

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OBJECTIVE

Graduate student transitioning to Data Science with 2+ years of experience analyzing system performance data and developing automation solutions. Proficient in Python, SQL, and data pipelines. Seeking Data Scientist/Analyst co-op roles to apply statistical modeling and machine learning to solve business problems.

EDUCATION

Master of Science in Information Systems, Northeastern University — Expected May 2027
GPA: [Your GPA]/4.0
Relevant Coursework: Database Management Systems, Data Analytics, Cloud Computing, System Design, Statistics

Bachelor of Technology in Electronics Engineering, Dwarkadas J. Sanghvi College of Engineering 2020 - 2023
Relevant Coursework: IoT Enterprise Networks, Operating Systems, Computer Networks, Data Structures, Algorithms

SKILLS

Languages & Libraries	Python, SQL, JavaScript, Java, C++, Pandas, NumPy, Scikit-learn, Matplotlib, Sea
Technologies/Framework	Git, Flask, Node.js, Express.js, Jupyter Notebook, Selenium, Postman, Docker
Data & Visualization	Tableau (learning), Excel, MongoDB, MySQL, Data Pipelines, ETL
Cloud & Tools	AWS (learning), Linux, CI/CD, JMeter, MQTT, REST APIs

EXPERIENCE

Software Engineer - Test Automation & Data Analysis — May 2023 - Aug 2024
UtopiaTech Pvt Ltd — *Mumbai, India*

- Analyzed device performance data from 10,000+ IoT users, identifying 3 critical failure patterns using Python and statistical analysis, reducing incident response time by 40% through predictive alerting system.
- Optimized testing workflows by examining calibration patterns across 50+ device types using Python automation, reducing manual testing time by 70% and improving accuracy by 15%.
- Evaluated backend system logs to diagnose and resolve 50+ critical bugs affecting platform stability, improving system uptime from 92% to 98% through data-driven debugging.
- Designed automated data collection pipeline processing 100K+ test records daily, enabling real-time monitoring dashboards and reducing reporting time from 4 hours to 15 minutes.
- Collaborated with cross-functional teams to define KPIs and implement A/B testing framework for feature releases, supporting data-driven product decisions affecting 10,000+ users.

Software Developer Intern - IoT Data Systems — Dec 2019 - May 2020
UtopiaTech Pvt Ltd — *Mumbai, India*

- Analyzed sensor data from 50+ IoT devices deployed across Smart City pilot project, identifying optimal calibration parameters that improved data accuracy by 25%.
- Optimized MongoDB queries by examining data access patterns and indexing strategies, improving database performance by 35% and reducing query response time from 800ms to 520ms.
- Developed Python data processing scripts to automate device performance analysis, reducing manual data validation time by 70% and enabling real-time anomaly detection.

- Created data visualization dashboards using Python to monitor device health metrics across multiple locations, enabling proactive maintenance and reducing device failures by 18%.

PROJECTS

Real-time IoT Data Pipeline & Analytics. Engineered end-to-end data pipeline processing sensor data from 100+ IoT devices using Python, MongoDB, and data validation techniques. Developed performance monitoring dashboard analyzing 50K+ daily records, identifying 3 key optimization opportunities that improved system efficiency by 22%. Implemented automated anomaly detection reducing false alerts by 40%. *Python, MongoDB, Pandas, Matplotlib*

MySpeech - Accessible Communication Platform. Built web conferencing platform with integrated ML-powered speech-to-text conversion serving 50+ concurrent users. Analyzed user engagement data to identify 3 key accessibility improvements, increasing user retention by 30%. Implemented real-time data processing using WebSocket and Node.js. Won Best Final Year Project Award. *Node.js, Express.js, MongoDB, ML APIs*

Predictive Maintenance System for IoT Devices. Developed Python-based security monitoring system analyzing access patterns from 3 residential buildings. Applied statistical methods to identify anomalous behavior, reducing unauthorized access by 80%. Created React Native mobile app for real-time alerts and data visualization. Integrated with cloud platform for remote monitoring. *Python, React Native, MQTT, Adafruit.io*

Automated Testing & Quality Analytics Framework. Designed comprehensive testing framework using Python and pytest for IoT device validation. Analyzed test results across 200+ device deployments to identify common failure modes, enabling targeted quality improvements. Implemented data-driven test prioritization reducing testing time by 45% while maintaining 95% coverage. *Python, Pytest, Selenium, Git*

EXTRA-CURRICULAR ACTIVITIES

- Building personal data science portfolio with 3+ projects on GitHub focusing on predictive analytics and data visualization at adityashah2811.github.io
- Pursuing Google Data Analytics Professional Certificate (Target: March 2025); completed 60% of coursework including Python for Data Science and Machine Learning
- Active member of Boston Data Science Meetup; attended 5+ sessions on ML model deployment, feature engineering, and A/B testing
- Contributing to open-source data analysis projects on GitHub; submitted 3 PRs for data pipeline improvements and automation tools

CERTIFICATIONS & LEARNING

- Google Data Analytics Professional Certificate (In Progress - 60% Complete, Target: March 2025)
- Python for Data Science and Machine Learning Bootcamp (Udemy, Completed 2024)
- AWS Cloud Practitioner (In Progress - Target: February 2025)